

Algebra 1: Unit 8, Lesson 9

1. A quadratic equation in standard form is shown.

$$y = \frac{1}{2}x^2 - 2x + 1$$

Sandra found the x -coordinate of the vertex. Her work is below.

$$x = \frac{-(-2)}{2\left(\frac{1}{2}\right)}$$

$$x = \frac{2}{1}$$

$$x = 2$$

The x -coordinate of the vertex is 2. What is the y -coordinate of the vertex?

2. A quadratic equation is shown.

$$y = 2(x + 6)^2 - 9$$

The equation is in vertex form. What is the vertex?

3. A photographer was attempting to get the best angle of a building from a helicopter. The height of the helicopter over time is represented by the equation, $H(x) = -1(x - 40)^2 + 1000$, where x is the time (in minutes) and H is the height of the helicopter (in feet).

Complete the table.

Key Feature	What it Represents in the Context
y -intercept	